

Comparing Fractions Without Being Fooled

Answer Key

Grades 4–5 Mathematics

Quick Answers

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| 1. 9 | 5. (a) lowest terms = $\frac{3}{5}$, — |
| 2. < | 6. (a) numerator of the first fraction in twelfths = 10, (b) correct comparison sign = >, — |
| 3. (a) numerator of the first fraction in fortieths = 15, (b) numerator of the second fraction in fortieths = 24, (c) correct comparison sign = <, — | 7. $\frac{3}{5}$, $\frac{2}{3}$, $\frac{7}{10}$ |
| 4. < | 8. (a) first comparison sign = >, (b) second comparison sign = <, — |
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What is verified

4 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 3, 5, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 4–5 Mathematics

4.NF.A – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

1. *If they got 11: added the same number to the top that was added to the bottom instead of multiplying*
 3. *If they got 3: replaced the denominator with 40 but left the numerator at 3*
 6. *If they got 5: changed only the denominator to 12 and left the numerator at 5*
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1. Fill in the missing numerator: $\frac{3}{4} = \frac{\quad}{12}$

Solution: The denominator went from 4 to 12, which is a multiplication by 3. To keep the same amount, the numerator must be multiplied by 3 as well: $3 \times 3 = 9$. 9

Common wrong answer: 11 — from adding 8 to the top because 8 was added to the bottom. Adding changes the amount; only multiplying by the same factor keeps it.

2. Write $<$ or $>$ between $\frac{4}{9}$ and $\frac{4}{7}$.

Solution: Both fractions count 4 pieces, so the only difference is how big a piece is. Ninths are smaller than sevenths, so four ninths is less than four sevenths.

$$\frac{4}{9} < \frac{4}{7}$$

3. Dev writes: “ $\frac{3}{8}$ is greater than $\frac{3}{5}$, because 8 is greater than 5.” (a) Rewrite $\frac{3}{8}$ over 40. (b) Rewrite $\frac{3}{5}$ over 40. (c) Correct sign. (d) Explain the error.

Solution (a): $8 \times 5 = 40$, so multiply the numerator by 5 too: 3×5 .

$$15$$

(b) $5 \times 8 = 40$, so multiply that numerator by 8: 3×8 .

$$24$$

(c) Over the same denominator the counts can finally be compared: 15 fortieths against 24 fortieths.

$$\frac{3}{8} < \frac{3}{5}$$

(d) — instructor-judged. Look for a response about piece *size*: when the numerators match, a larger denominator means the whole was cut into more and therefore smaller pieces, so three of them is less, not more. Dev compared the denominators as if they were ordinary whole numbers, which reverses the truth.

Common wrong answer: 3 in part (a) — swapping the denominator for 40 while leaving the numerator alone, which is the same denominator-only habit in a different disguise.

4. Use one half as a benchmark. Write $<$ or $>$ between $\frac{2}{7}$ and $\frac{5}{8}$.

Solution: Half of 7 is 3.5, and 2 is less than that, so $\frac{2}{7}$ is below one half. Half of 8 is 4, and 5 is more than that, so $\frac{5}{8}$ is above one half. One fraction below the benchmark and one above settles it with no common denominator at all.

$$\frac{2}{7} < \frac{5}{8}$$

5. Lena writes: “ $\frac{6}{10}$ and $\frac{3}{5}$ must be different amounts, because none of their digits match.”

(a) Write $\frac{6}{10}$ in lowest terms. (b) Are they equivalent? Explain what Lena missed.

Solution (a): 6 and 10 share the factor 2. Dividing both by 2 gives

$$\frac{3}{5}$$

(b) — instructor-judged. Yes, they are equivalent. Look for a response that names the common factor: dividing the top and the bottom by the same number, 2, regroups the pieces without changing how much of the whole is covered. Six tenths and three fifths shade the same strip. A response that only asserts “they are the same” without pointing at the shared factor is incomplete.

6. Sam writes: “ $\frac{5}{6}$ is less than $\frac{7}{12}$, because 5 is less than 7.” (a) Rewrite $\frac{5}{6}$ over 12. (b) Correct sign. (c) Explain the error.

Solution (a): $6 \times 2 = 12$, so the numerator doubles too: 5×2 .

$$10$$

(b) Now both are twelfths: 10 twelfths against 7 twelfths.

$$\frac{5}{6} > \frac{7}{12}$$

(c) — **instructor-judged.** Look for a response that says a numerator only counts pieces and says nothing about their size, so 7 small pieces can be less than 5 large ones. Numerators may be compared only once both fractions are written over the same denominator.

Common wrong answer: 5 in part (a) — changing the denominator to 12 and leaving the numerator untouched.

7. Put $\frac{3}{5}$, $\frac{7}{10}$ and $\frac{2}{3}$ in order from least to greatest.

Solution: All three sit above one half, so the benchmark alone will not separate them. Use the common denominator 30: $\frac{3}{5} = \frac{18}{30}$, $\frac{7}{10} = \frac{21}{30}$, $\frac{2}{3} = \frac{20}{30}$. Ordering the counts 18, 20, 21 gives the order below.

$\frac{3}{5} < \frac{2}{3} < \frac{7}{10}$
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8. Priya claims: “When two fractions have the same numerator, the one with the bigger denominator is the bigger fraction.” (a) Sign between $\frac{6}{7}$ and $\frac{6}{13}$. (b) Sign between $\frac{5}{12}$ and $\frac{5}{8}$. (c) Write the rule that is actually true, and explain why.

Solution (a): Sevenths are bigger pieces than thirteenths, and both fractions count 6 pieces.

$\frac{6}{7} > \frac{6}{13}$

(b) Twelfths are smaller pieces than eighths, and both count 5.

$\frac{5}{12} < \frac{5}{8}$

(c) — **instructor-judged.** The true rule is the reverse of Priya’s: with the same numerator, the *smaller* denominator gives the larger fraction. Look for a reason built on piece size — a bigger denominator splits the whole into more pieces, so each piece is smaller, and the same count of smaller pieces adds up to less. Both the rule and the reason are needed for full credit; the corrected rule alone is half an answer.